

Lab Report Summary

- Kali + Nessus Essentials
- Metasploitable 2
- Ollama AI API
- Local Lab Only

What was done

- Performed host discovery on the lab subnet 10.183.55.0/24.
- Ran Basic Network Scan against Metasploitable 2 at 10.183.55.243.
- Configured a vulnerable AI/LLM lab by exposing Ollama on 10.183.55.37:11434.
- Used Nessus Find AI scan to detect AI/LLM exposure and validate the finding.

Key evidence from screenshots

- Nmap identified Metasploitable services: FTP, SSH, Telnet, HTTP, SMB, MySQL, PostgreSQL, VNC and others.
- Host Discovery in Nessus found 4 live hosts in the lab network.
- Find AI scan detected Ollama Unauthenticated Access as Critical with CVSS 10.0.
- Basic Network Scan found 71 vulnerabilities on Metasploitable 2.

Why this project is useful

- Shows practical Nessus workflow: discovery -> vulnerability scan -> finding review.
- Adds current AI/LLM attack surface awareness using a real exposed AI runtime.
- Evidence is documented with screenshots because Nessus Essentials export is limited.

```
(root@kali)-[/home/kali/posti-tnt-js]
# nmap 10.183.55.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-13 06:32 -0400
Nmap scan report for 10.183.55.68
Host is up (0.0024s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 02:2F:87:5F:53:AA (Unknown)

Nmap scan report for 10.183.55.203
Host is up (0.00022s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
1234/tcp  open  hotline
MAC Address: DC:46:28:33:61:ED (Intel Corporate)

Nmap scan report for 10.183.55.243
Host is up (0.00041s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:05:26:93 (VMware)

Nmap scan report for 10.183.55.37
Host is up (0.00094s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
```

Step 1 - Host Discovery and Target Identification

Subnet scan: 10.183.55.0/24

host discovery

Configure

Audit Trail

Launch

Report

Export

Back to My Scans

Hosts4

Vulnerabilities2

History1

Filter

Search Hosts

4 Hosts

☐	Host	Ports	
☐	10.183.55.243	111, 139, 445, 2049, 34257, 34852, 38347, 39639, 58336, 58907	✗
☐	10.183.55.203		✗
☐	10.183.55.68		✗
☐	10.183.55.37		✗

Scan Details

Policy:Host Discovery

Status:Completed

Severity Base:CVSS v3.0

Scanner:Local Scanner

Start:Today at 7:15 AM

End:Today at 7:18 AM

Elapsed:3 minutes

Vulnerabilities

Critical

High

Medium

Low

Info

The screenshot displays the Tenable Nessus Essentials web interface. At the top, a navigation bar shows the current path: Home > Scans > Settings. The main heading is 'New Scan / Host Discovery'. Below this, there are tabs for 'Settings' and 'Plugins'. The 'Settings' tab is active, showing a configuration form for a new scan. The form has a sidebar with categories: BASIC (selected), SCHEDULE, DISCOVERY, REPORT, and ADVANCED. Under the 'BASIC' category, the following fields are visible:

- Name:** host discovery
- Description:** (empty text area)
- Folder:** My Scans (dropdown menu)
- Targets:** 10.181.55.0/24 (text area)
- Upload Targets:** (button)
- Add File:** (button)

 A 'Tenable News' notification is present in the bottom left corner, stating 'The Armitage battle is already happening. Watch it...'. The bottom status bar shows the current session information: 'To direct input to this VM, move the mouse pointer inside or press Ctrl+G'.

What I did

- Created a Nessus Host Discovery scan named "host discovery".
- Target range: 10.183.55.0/24.
- Goal: identify live machines before detailed vulnerability scans.

Result interpretation

- 4 live hosts discovered.
- 10.183.55.243: Metasploitable 2 with many exposed ports.
- 10.183.55.37: Kali/Ollama AI target.
- 10.183.55.203: AI/API host with port 1234 seen in Nmap.
- 10.183.55.68: DNS service on port 53.

Step 2 - Basic Network Scan on Metasploitable 2

Target: 10.183.55.243

Restore Session

Install Tenable Nessus on

Download Tenable Nessu

Nessus Essentials / Folde

GitHub

127.0.0.1:8834/#/scans/reports/9/hosts/2/vulnerabilities

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB LinkFinder Output

tenable Nessus Essentials Scans Settings

FOLDERS

My Scans

test

All Scans

Trash

RESOURCES

Policies

Plugin Rules

Tenable News

The AI-vs-AI battle is already happening. Watch it...

Read More

basic network scan / 10.183.55.243

Back to Hosts

Vulnerabilities 71

Filter Search Vulnerabilities 71 Vulnerabilities

Sev	CVSS	VPR	EPSS	Name	Family	Count	
CRITICAL	10.0 *	7.4	0.8663	UnrealIRCd Backdoor Detection	Backdoors	1	
CRITICAL	10.0			Canonical Ubuntu Linux SEoL (8.04.x)	General	1	
CRITICAL	10.0 *			VNC Server 'password' Password	Gain a shell remotely	1	
CRITICAL	9.8	8.9	0.9447	Apache Tomcat AJP Connector Request Injection (Ghostcat)	Web Servers	1	
CRITICAL	9.8			SSL Version 2 and 3 Protocol Detection	Service detection	2	
CRITICAL	9.8			Bind Shell Backdoor Detection	Backdoors	1	
CRITICAL	...	...	...	SSL (Multiple Issues)	Gain a shell remotely	3	
HIGH	7.5 *	6.7	0.5006	rsh Service Detection	Service detection	1	
HIGH	7.5	5.9	0.7852	Samba Badlock Vulnerability	General	1	
HIGH	7.5			NFS Shares World Readable	RPC	1	
MIXED	...	...	...	SSL (Multiple Issues)	General	28	
MIXED	...	...	...	ISC Bind (Multiple Issues)	DNS	5	
MEDIUM	6.5			TLS Version 1.0 Protocol Detection	Service detection	2	
MEDIUM	6.5			Unencrypted Telnet Server	Misc.	1	Modify

Host Details

IP: 10.183.55.243

MAC: 00:0C:29:05:26:93

OS: Linux Kernel 2.6 on Ubuntu 8.04 (hardy)

Start: Today at 7:18 AM

End: Today at 7:33 AM

Elapsed: 15 minutes

KB: Download

Auth: Fail

Vulnerabilities

Critical

High

Medium

Low

Info

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Screenshot details

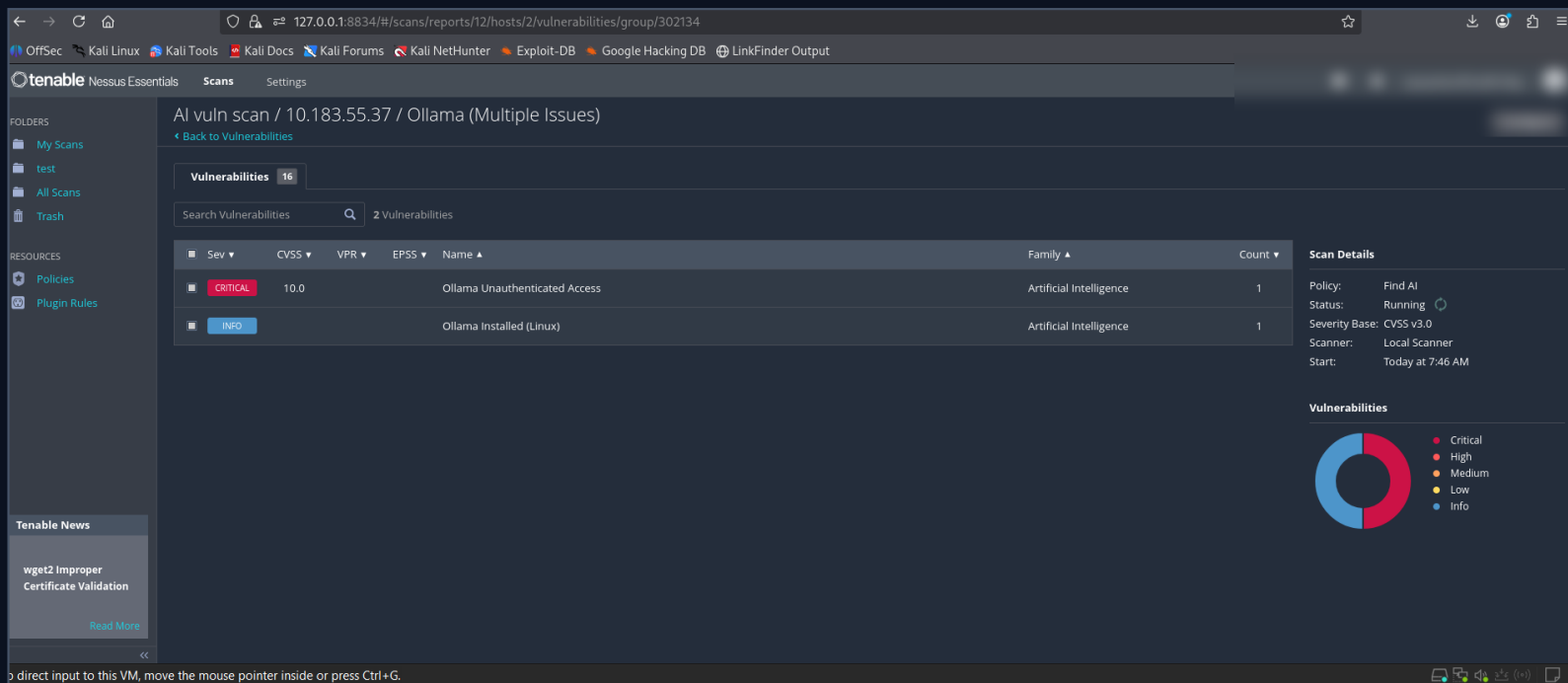
- Nessus Basic Network Scan results show 71 vulnerabilities for host 10.183.55.243.
- Top critical items visible: UnrealIRCd Backdoor Detection, VNC password "password", Apache Tomcat AJP Ghostcat, SSLv2/v3, Bind Shell Backdoor.
- The host is identified as Linux kernel 2.6 on Ubuntu 8.04, matching Metasploitable 2 lab behavior.

Security learning

- This scan represents traditional infrastructure VAPT: old services, backdoors, weak credentials, legacy SSL/TLS and exposed network services.
- Priority should start with remote shell/backdoor findings and default credential issues before lower-impact informational findings.

Step 3 - AI/LLM Exposure Scan using Ollama

Target: 10.183.55.37:11434



AI vuln scan / 10.183.55.37 / Ollama (Multiple Issues)

Back to Vulnerabilities

Vulnerabilities 16

Search Vulnerabilities 2 Vulnerabilities

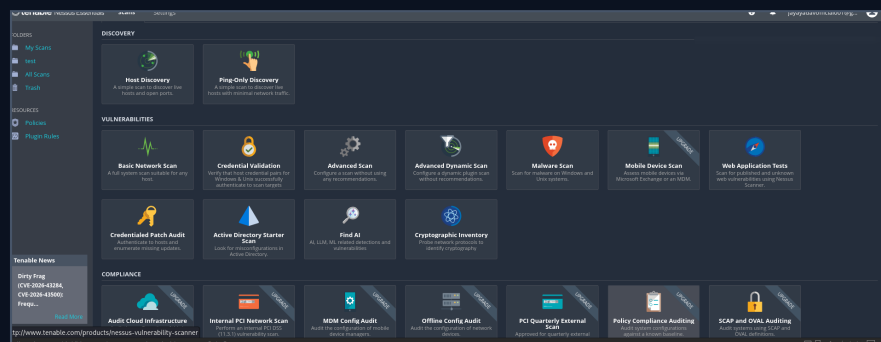
Sev	CVSS	VPR	EPSS	Name	Family	Count
CRITICAL	10.0			Ollama Unauthenticated Access	Artificial Intelligence	1
INFO				Ollama Installed (Linux)	Artificial Intelligence	1

Scan Details

Policy: Find AI
Status: Running
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 7:46 AM

Vulnerabilities

Donut chart showing severity distribution: Critical (red), High (orange), Medium (yellow), Low (green), Info (blue).



What I did here

- Selected Nessus "Find AI" template.
- This scan checks AI, LLM and ML related detections/vulnerabilities.
- Ollama was intentionally exposed on 0.0.0.0:11434 for lab testing.

Finding captured

- Critical: Ollama Unauthenticated Access.
- CVSS: 10.0.
- Family: Artificial Intelligence.
- Info: Ollama Installed (Linux).
- Risk: unauthenticated users on the network can reach the AI runtime/API and consume compute resources.

Remediation summary

- Metasploitable-style findings: patch outdated services, disable unused ports, remove default credentials, replace insecure protocols and restrict internal services.
- Backdoor/remote shell findings: treat as immediate critical risk, isolate host, investigate compromise indicators and rebuild from trusted image.
- Ollama exposure: bind to 127.0.0.1, add firewall restrictions, place behind authenticated reverse proxy if remote access is required, and monitor API usage.

GitHub evidence to upload

- screenshots/host-discovery-config.png
- screenshots/host-discovery-results.png
- screenshots/metasploitable-basic-network-scan.png
- screenshots/find-ai-template.png
- screenshots/ollama-critical-finding.png
- notes/findings-summary.md and notes/commands-used.md

Project outcome

Area	Evidence	Result
Discovery	Nessus Host Discovery + Nmap	Identified 4 lab hosts and target roles
Traditional VAPT	Basic Network Scan on Metasploitable 2	71 vulnerabilities, including critical remote access/backdoor issues
AI/LLM VAPT	Find AI scan on exposed Ollama	Critical Ollama Unauthenticated Access, CVSS 10.0
Documentation	Screenshots + manual notes	Created evidence package despite Nessus Essentials export limitation

Conclusion: This lab demonstrates a complete scanner-based workflow: discover hosts, scan a vulnerable infrastructure target, then assess a modern AI/LLM runtime exposure. The AI finding is especially useful for a fresher project because it shows awareness of emerging attack surface beyond classic services.